OpenGeoSys Training – KIGAM/ZJU 2026

Exercise 0: Overview and OGS installation

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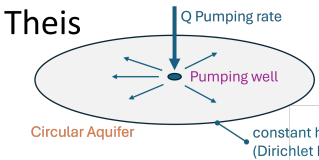
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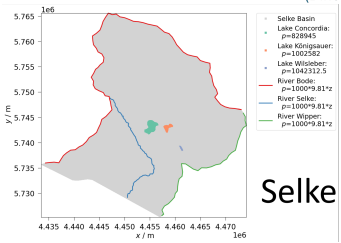
Daejeon/Hangzhou, 14th October 2026

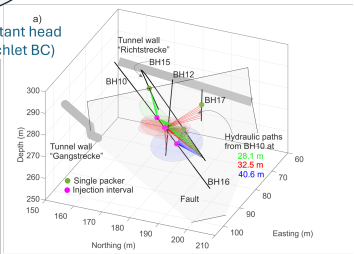
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Agenda

1. OGS Course Overview
2. OGS exercises
   - Theis: wellbore hydraulics
   - STIMTEC: hydraulic test analysis
3. OGS installation
4. OGS course repo
5. References







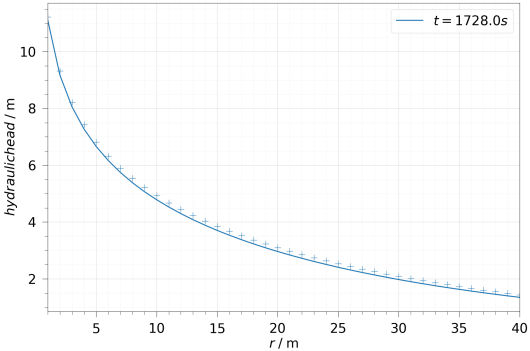
OGS exercises

- Theis: wellbore hydraulics
- STIMTEC: hydraulic test analysis in fractured rock (URL)
- Selke: catchment analysis

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Contents

- Analytical solution: Well function $W(u)$
- Comparison with numerical solution (H process basic model)
- Additional processes
  - Leakage through fracture boundaries
  - Leakage through rock matrix
- Homework: Building your own Jupyter Notebook
- OGS reference: Benchmark gallery

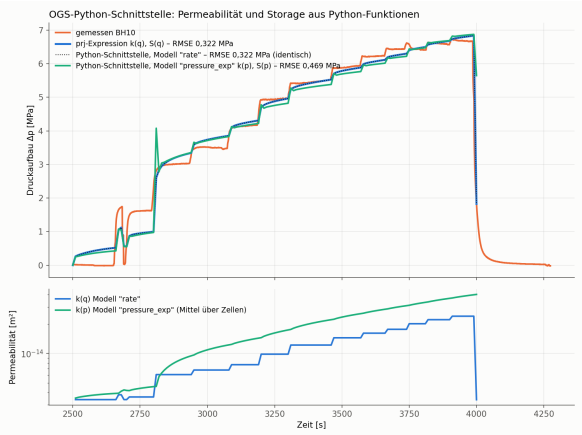


STIMTEC: hydraulic test analysis in fractured rock (URL

Exercise overview

Contents

- Two pumping tests in URL ”Reiche Zeche” (BH10_40 and BH10_51)
- Modeling strategy:
  - Closed fracture
  - Leakage
  - Pressure-dependent hydraulic properties
  - Single fracture vs. discrete fracture network (DFN)
- **AI use for model extension**
- OGS reference: Benchmark gallery



Course repo's

- Olaf: Hydraulics (H)
- Mehran: Coupled processes (THM)

GitHub: `https://github.com/OlafKolditz/ogs-course-2026`

short `git` introduction

- install `git`
- `git clone repo`
- `git clone https://github.com/OlafKolditz/ogs-course-2026`
- `git fetch`
- `git pull`

see tutorial:
`https://www.overleaf.com/9159116795bmhsfhgtbmtb#6eb5a5`

References

[1] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. ISBN: 9783662047613. DOI: 10.1007/978-3-662-04761-3. URL: http://dx.doi.org/10.1007/978-3-662-04761-3.

[2] Erik Nixdorf, Mike Sips and Peter Morstein. 'Development of a digital workflow for layer reduction strategies and visual-analytical outcome analysis to enhance geo-hydraulic modelling efficiency'. In: Digital Water 3.1 (2025), pp. 1–24. DOI: 10.1080/28375807.2025.2460825.

[3] Charles V. Theis. 'The Relation Between the Lowering of the Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage'. In: Transactions, American Geophysical Union 16.2 (1935), pp. 519–524. DOI: 10.1029/TR016i002p00519.

[4] Zhi-Yuan Zhang, Christian Schmidt, Erik Nixdorf, Xingxing Kuang and Jan H. Fleckenstein. 'Effects of Heterogeneous Stream-Groundwater Exchange on the Source Composition of Stream Discharge and Solute Load'. In: Water Resources Research 57.8 (Aug. 2021), e2020WR029079. DOI: 10.1029/2020WR029079.